

MODULE 07

Ankle & Foot MRI: Ligaments, Tendons & Fracture Classification

ATFL Grading · Sanders CT · Achilles · Peroneal Tendons · Charcot · Turf Toe · Freiberg

1.0 AMA PRA Cat 1

Ankle

CME Article Review

LEARNING OBJECTIVES

1. Grade lateral ankle ligament injuries (ATFL, CFL, PTFL) on fluid-sensitive MRI.
2. Apply Sanders CT classification to calcaneal fractures (I–IV).
3. Identify and classify peroneal tendon pathology on axial MRI.
4. Recognize Achilles tendinopathy, partial, and complete tears on sagittal MRI.
5. Apply Eichenholtz staging to Charcot neuroarthropathy (stages 0–III).
6. Classify turf toe and Freiberg infraction using established grading systems.

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1. Ankle Lateral Ligament Injuries & Key Measurements

Lateral Ligament Grading (Fluid-Sensitive MRI)

Grade	Pathology	MRI Findings	Clinical
I — Sprain	Intact fibers	Periligamentous edema; thickened; continuous	Tender; stable
II — Partial	Partial disruption	Partial discontinuity; T2 within ligament	Partial laxity
III — Complete	Full-thickness rupture	Absent/wavy ligament; fluid gap	Instability; positive drawer

Key Ankle Measurements

Measurement	Normal Value	Significance
ATFL thickness (axial)	<3 mm non-injured	Thickening >3 mm = chronic tear/fibrosis
Tibiotalar medial space (coronal)	2–4 mm	Asymmetry >1 mm = mortise tilt
Syndesmotic width	≤5 mm (anterior tibiofibular)	Widening = syndesmotic injury
Achilles AP diameter (sagittal)	<6 mm	≥6 mm = tendinopathy threshold

2. Sanders Calcaneal Fracture Classification (Coronal CT)

Type	Fracture Lines	Posterior Facet Fragments	Management
I	Non-displaced (any pattern)	Non-displaced	Non-operative
II	One vertical line	2 fragments (A/B/C subtypes)	ORIF — good outcomes
III	Two vertical lines	3 fragments (AB, AC, BC subtypes)	ORIF — complex; fair outcomes
IV	Comminuted; ≥3 lines	>3 fragments	Primary subtalar fusion or non-op

3. Tendons, Charcot & Forefoot

Achilles Tendon MRI Classification

Finding	AP Diameter	MRI Appearance	Significance
Normal	<6 mm	Uniform low T1/T2; parallel fibers	No pathology

Finding	AP Diameter	MRI Appearance	Significance
Tendinopathy	6–10 mm	Fusiform thickening; intratendinous T2	Non-surgical; eccentric loading
Partial tear	Variable	Focal high T2; partial fiber discontinuity	Consider surgical vs. conservative
Complete rupture	N/A (gap)	Full-thickness gap; tendon ends retracted	Surgical repair or boot

Eichenholtz Staging — Charcot Neuroarthropathy

Stage	Name	Radiograph	Management
0 (Sanders)	Prodromal	Normal; clinical swelling/warmth	TCC; absolute offloading
I	Fragmentation	Cortical disruption; fracture; subluxation; debris	TCC; ABSOLUTE offloading
II	Coalescence	Decreased debris; periosteal new bone; early repair	Continue offloading; boot transition
III	Reconstruction	Remodeling; residual deformity; stable	Custom footwear; surgical reconstruction if unstable

CRITICAL

MRI CANNOT reliably distinguish Stage I Charcot from osteomyelitis — both show BME and soft tissue changes. Clinical context (neuropathy, ulcer, temperature gradient), lab markers (ESR/CRP/WBC), and biopsy may be required.

Turf Toe Grading (1st MTP Plantar Plate)

Grade	Pathology	MRI
I	Micro-tear; sprain	Periarticular edema; intact structures
II	Partial plantar plate tear	Partial disruption; edema
III	Complete disruption ± sesamoid Fx ± dislocation	Full-thickness tear; sesamoid retraction

References

- Sanders R et al. J Trauma 1993.
- Eichenholtz SN. Charcot Joints. Thomas 1966.
- Anderson RB, Hunt KJ. Sports Health 2010.